

Enterprise AI with Azure

3-Week Instructor-Led Training Program

Delivered by Learnlytica | Enterprise AI Learning & Skilling Platform

Duration 3 Weeks / 15 Days	Format Instructor-Led + Hands-On Labs	Technology Stack .NET · React · Azure AI	Hands-On Labs 15 Real-World Scenario Labs
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About This Program

This three-week program takes participants from AI fundamentals to building production-ready enterprise AI systems on Azure. The curriculum is designed for full-stack developers (.NET + React) who need to integrate Machine Learning, Azure AI cognitive services, Generative AI, and intelligent agents into real enterprise applications.

Every day is anchored to a real-world business scenario — a manufacturing plant, a financial services firm, a retail chain, a healthcare provider, or a logistics company — so participants immediately see the enterprise relevance of every technique they learn. Labs are hands-on, scenario-driven, and designed to produce deployable artefacts.

Program Objectives

- By the end of this program, participants will be able to:
- Explain the AI/ML/GenAI landscape and select the right approach for enterprise use cases
 - Build, train, evaluate, and deploy ML models on Azure Machine Learning
 - Integrate Azure AI services (OCR, Vision, Speech) into .NET and React applications
 - Design and implement Retrieval-Augmented Generation (RAG) pipelines for enterprise document Q&A
 - Apply structured prompt engineering to control LLM output quality and reliability
 - Build and orchestrate multi-step AI agents with function calling and memory
 - Create enterprise copilots using Copilot Studio and connect them to real data sources
 - Apply responsible AI principles: bias mitigation, content filtering, and cost governance
 - Deliver a complete end-to-end AI-powered enterprise application as a capstone project

Learning Approach: Scenario-First Methodology

Every session follows a scenario-first structure. Participants are presented with a real enterprise problem at the start of each day — drawn from industries such as retail, banking, manufacturing, healthcare, and logistics. All concepts, code, and labs are framed around solving that problem.

Step	Component	How It Works
1	Scenario Brief	A real enterprise problem is introduced: business context, pain points, and the AI solution to be built. This frames every concept that follows.

Step	Component	How It Works
2	Concept Session	Trainer-led walkthrough of the relevant Azure AI capabilities, architectural patterns, and implementation considerations.
3	Guided Lab	Participants implement a working solution in a live Azure environment. Labs use realistic datasets (invoices, customer records, product catalogues, audio files) — not toy examples.
4	Integration Check	Code is integrated with the .NET backend and React frontend so participants see the full system, not just an isolated API call.

Week 1: Foundations — ML Essentials, Azure Basics & Full-Stack Integration

Week 1 establishes the foundation: understanding the AI/ML landscape, building and evaluating ML models, deploying them on Azure Machine Learning, and integrating predictions into .NET + React applications.

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 1 <i>AI Overview + Architecture + Python</i>	- AI vs ML vs Deep Learning vs GenAI taxonomy - Enterprise AI use case mapping .NET + React + Azure end-to-end architecture - Python + Pandas: data loading & cleaning - Azure resource groups, storage accounts, blob storage	 Scenario: <i>RetailCo wants to understand which AI technology applies to their customer churn, demand forecasting, and chatbot requirements</i> Lab: Enterprise AI Architecture Mapping Participants map 6 real RetailCo business problems to the correct AI technology tier (rules engine, classical ML, GenAI). They set up an Azure resource group, create blob storage, and upload a sample customer dataset.	Full day ~7 hrs
Day 2 <i>ML Essentials</i>	- Supervised vs Unsupervised learning - Linear vs Non-linear model selection - Overfitting / Underfitting intuition - Bias-Variance tradeoff - Evaluation metrics: accuracy, precision, recall, F1	 Scenario: <i>A telecom company is losing 18% of customers per quarter and needs to predict who will churn next month</i> Lab: Customer Churn Prediction Model Using a 50,000-row telecom dataset (contract type, usage patterns, support tickets), participants train a Random Forest classifier, compare it against Logistic Regression, analyse the confusion matrix, and identify the top 5 churn-driving features.	Full day ~7 hrs
Day 3 <i>ML Workflow + Azure ML + Integration</i>	- Data preprocessing: missing values, encoding, scaling - Train-test split and cross-validation - Azure ML workspace, compute clusters, endpoints	 Scenario: <i>A retail bank needs to deploy its fraud detection model to production so the transactions API can call it in real time</i> Lab: Fraud Detection API — Cloud Deployment Participants take a trained fraud detection model, register it in Azure ML, deploy it as a managed online endpoint, call it from a	Full day ~7 hrs

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 4 Azure AI Services + Blob Storage	- Deploying a model as a REST endpoint - Consuming ML endpoint from .NET and React	.NET Web API, and display the risk score on a React transaction dashboard. End-to-end integration tested with live HTTP calls.	
	- Azure AI services portfolio overview - Blob storage: upload, download, SAS tokens - Managing API keys and service endpoints - Connecting storage to AI service pipelines - Backend orchestration patterns in .NET	 Scenario: A logistics company receives 200 shipment documents per day and needs a storage and processing backbone before adding AI extraction Lab: Document Intake Pipeline Setup Participants build a document intake pipeline: React drag-and-drop upload → .NET API → Azure Blob Storage. They set up Azure AI service credentials, configure access policies, and verify the end-to-end document flow that will power the next day's OCR lab.	Full day ~7 hrs
	- Document Intelligence: invoice and form extraction - Extracting structured fields from PDFs and images - Computer Vision: tags, captions, object detection - Parsing JSON responses in .NET - Rendering extracted data in React UI	 Scenario: The same logistics company now needs to automatically extract PO numbers, vendor names, amounts, and line items from incoming PDF invoices — eliminating manual data entry Lab: Automated Invoice Processing System Using a set of 10 realistic invoice PDFs (varying layouts, multi-page, handwritten fields), participants call Document Intelligence, parse the structured response in .NET, handle low-confidence extractions with a review flag, and render the extracted fields in a React form with edit capability.	Full day ~7 hrs

Week 1 Lab Outcomes: Churn prediction model • Fraud detection API on Azure ML • Document intake pipeline • Invoice extraction system

Week 2: Generative AI + RAG — LLMs, Prompt Engineering & Knowledge Systems

Week 2 moves into Generative AI: understanding LLMs, building reliable chatbots with structured prompting, creating semantic search systems, and implementing RAG pipelines for enterprise document Q&A.

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 6 Speech-to-Text + Integration	- Speech recognition: real-time vs batch - Audio capture from React frontend - Sending audio streams to .NET backend - Azure Speech SDK integration - Displaying and storing transcripts	 Scenario: A healthcare provider needs to let doctors dictate patient notes hands-free during consultations and automatically save them to the patient record system Lab: Clinical Note Dictation System Participants build a voice-capture widget in React (MediaRecorder API), stream audio to a .NET endpoint, call Azure Speech-to-Text, post-process the transcript for medical terminology, and display the	Full day ~7 hrs

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 7 <i>Generative AI Basics + Azure OpenAI</i>	- LLM architecture: tokens, context window, temperature - Azure OpenAI service setup and model deployment - Chat completions API: system, user, assistant roles - Building a basic chatbot in .NET - Streaming responses to React UI	formatted note. Tested with sample doctor dictation recordings.  Scenario: <i>An e-commerce company wants a customer support chatbot that answers product questions, order status queries, and return policy questions — without routing to a human agent</i> Lab: E-Commerce Support Chatbot (v1) Participants configure Azure OpenAI, design a system prompt that enforces the assistant's persona and scope, implement chat history management in .NET, stream responses to a React chat UI, and test edge cases where the bot should decline to answer. Token usage monitoring included.	Full day ~7 hrs
Day 8 <i>Prompt Engineering</i>	- System vs user vs assistant prompt roles - Few-shot prompting with real examples - Structured output: JSON schemas via prompting - Chain-of-thought and step-back prompting - Prompt debugging and iterative refinement	 Scenario: <i>A financial services firm needs the LLM to generate standardised credit assessment summaries from unstructured analyst notes — the output must always follow a fixed JSON schema for downstream processing</i> Lab: Structured Credit Assessment Generator Participants take 8 raw analyst note samples (varying quality and style) and engineer a prompt system that reliably extracts: risk rating, key positives, key negatives, and recommendation — all as valid JSON. They measure output consistency across samples and handle schema violation cases with retry logic.	Full day ~7 hrs
Day 9 <i>Embeddings + Semantic Search</i>	- What are embeddings and why they matter - Cosine similarity and vector space intuition - Azure AI Search: indexing and querying - Hybrid search: semantic + keyword - Chunking strategies for long documents	 Scenario: <i>A professional services firm has 5,000 internal policy documents, contracts, and procedure guides. Employees waste 2 hrs/day searching for the right document — keyword search fails on jargon-heavy content</i> Lab: Enterprise Knowledge Search System Participants chunk and embed a corpus of 30 sample policy documents, index them in Azure AI Search, implement a hybrid semantic + keyword search endpoint in .NET, and build a React search UI that returns the top 5 relevant chunks with source attribution and relevance scores.	Full day ~7 hrs
Day 10 <i>RAG System + Evaluation</i>	- RAG architecture: retrieval → augmentation → generation - Prompt orchestration: injecting retrieved context	 Scenario: <i>The same professional services firm now needs employees to ask natural language questions and get accurate, sourced answers from the internal knowledge base — not hallucinated responses</i>	Full day ~7 hrs

Day	Topics Covered	Real-World Scenario & Lab	Duration
	- Hallucination detection strategies - Evaluation metrics: faithfulness, relevance, answer quality - Grounding responses with source citations	Lab: Policy Q&A System with Grounded Answers Participants build the full RAG pipeline on top of Day 9's search index: question → retrieval (top 3 chunks) → prompt construction → Azure OpenAI generation → answer with citations. They test 15 adversarial questions designed to trigger hallucinations and implement a confidence scoring mechanism that flags low-grounding responses for human review.	

Week 2 Lab Outcomes: Clinical dictation system • E-commerce support chatbot • Structured JSON generator • Semantic search engine • RAG Q&A pipeline with grounding

Week 3: Agents + Copilot Studio + Responsible AI + Capstone

Week 3 advances to autonomous AI agents, multi-step orchestration, Copilot Studio enterprise bots, and production-grade responsible AI practices — culminating in a full-scale capstone project.

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 11 <i>Agent Fundamentals</i>	- Agents vs chatbots: capability comparison - Tools and function definitions - Agent decision loops and reasoning traces - When to use agents vs single-shot LLM calls - Azure OpenAI function calling basics	 Scenario: An IT helpdesk receives 300 tickets per day. 60% are routine: password resets, software installs, VPN access requests. The company wants an AI agent that resolves tier-1 tickets autonomously Lab: IT Helpdesk Tier-1 Support Agent Participants implement a support agent with 4 tools: check_ticket_status, reset_password, check_software_catalogue, and escalate_to_human. The agent processes 10 sample helpdesk tickets, reasoning through which tool to call and when to escalate. Participants observe and validate the agent's decision trace.	Full day ~7 hrs
Day 12 <i>Agent Implementation</i>	- Function calling: schema design and response parsing - Multi-step tool chaining and state passing - Conversation memory: short-term and summary memory - Error handling: tool failures and retries - Logging agent actions for auditability	 Scenario: A procurement team needs an agent that can look up vendor catalogues, check inventory levels, raise purchase orders, and notify the approver — a multi-step workflow currently done manually across 3 systems Lab: Procurement Workflow Automation Agent Participants build a 6-tool procurement agent (search_vendor, check_inventory, get_pricing, create_po, send_approval_email, check_po_status). The agent handles a realistic 8-step procurement scenario: from receiving a purchase request to confirming PO creation. Memory ensures context is retained across the workflow.	Full day ~7 hrs

Day	Topics Covered	Real-World Scenario & Lab	Duration
Day 13 <i>Advanced Agents</i>	- Planning agents vs reactive agents - Multi-tool orchestration and parallel calls - State machine patterns for long-running workflows - Error recovery and compensating transactions - Monitoring agent performance and cost	 Scenario: <i>A financial services operations team needs an autonomous reconciliation agent that checks daily transactions, identifies mismatches, queries the source system for corrections, and generates a daily exceptions report</i> Lab: Autonomous Financial Reconciliation Agent Participants build a planning-based reconciliation agent that ingests a daily transaction file (200 rows), cross-references it against a mock ledger API, categorises discrepancies (timing, amount, missing), calls a correction API for auto-resolvable cases, and generates a structured exceptions report. The agent handles API failures with retry logic.	Full day ~7 hrs
Day 14 <i>Copilot Studio + Responsible AI</i>	- Copilot Studio: declarative agent authoring - Connecting to enterprise data sources and APIs - Extending copilots with custom plugins - Responsible AI: bias, fairness, privacy by design - Content filtering, guardrails, cost monitoring	 Scenario: <i>An HR department wants an employee self-service copilot that answers leave policy questions, checks payslip data, initiates onboarding workflows, and routes complex queries to HR — all within Microsoft Teams</i> Lab: HR Self-Service Copilot (Teams-integrated) Participants build a Copilot Studio agent connected to a mock HR API (leave balances, policy docs, payslip data). They configure content safety filters to prevent inappropriate responses, set up topic-level fallback to a human agent, and test 12 HR query scenarios including edge cases with sensitive personal data requests.	Full day ~7 hrs
Day 15 <i>Capstone: End-to-End AI Enterprise Application</i>	- Full system architecture design - Integrating ML + Vision/OCR + GenAI + Agents .NET backend orchestration layer - React frontend with unified AI UX - Testing, deployment, and production handoff	 Scenario: <i>GlobalRetail Ltd needs an AI-powered store operations assistant: sales forecasting (ML), supplier invoice processing (OCR), inventory Q&A from policy documents (RAG), and an autonomous reorder agent — all accessible from a single React dashboard</i> Lab: GlobalRetail AI Operations Platform — Capstone Teams of 2–3 participants build and integrate four system components: (1) a deployed Azure ML sales forecast endpoint feeding a React chart, (2) an invoice OCR pipeline for supplier documents, (3) a RAG Q&A interface over inventory policies, and (4) an autonomous reorder agent that triggers purchase orders when stock thresholds are breached. Final demo to the group.	Full day ~7 hrs

Week 3 Lab Outcomes: IT helpdesk agent • Procurement automation agent • Financial reconciliation agent • HR copilot in Teams • Full-stack AI enterprise capstone

What Participants Receive

Learning Materials	Reusable Code Assets	Assessment & Recognition
• Student guides for all 15 days • Lab instruction sheets and solution code • Architecture reference diagrams • Azure resource setup checklists	• 15 working lab repositories (.NET + React) • Prompt library from prompt engineering session • RAG pipeline template • Agent orchestration starter kit	• Daily scenario-based knowledge checks • Capstone project evaluation rubric • Learnlytica completion certificate • Digital credential on program completion

Prerequisites

Recommended Knowledge	Technical Setup Required
• Working knowledge of C# / .NET Web API development • Basic React and JavaScript familiarity • Foundational Python and data handling (Pandas) • No prior AI/ML experience required	• Azure subscription with Contributor access • Visual Studio 2022 + VS Code installed • Node.js 18+ and .NET 8 SDK • Azure CLI and access credentials shared before Day 1

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